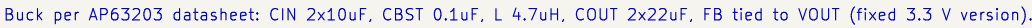
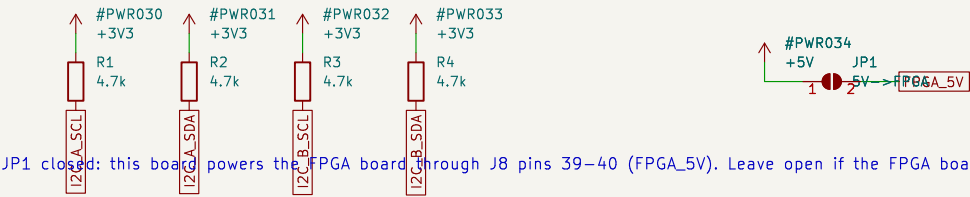
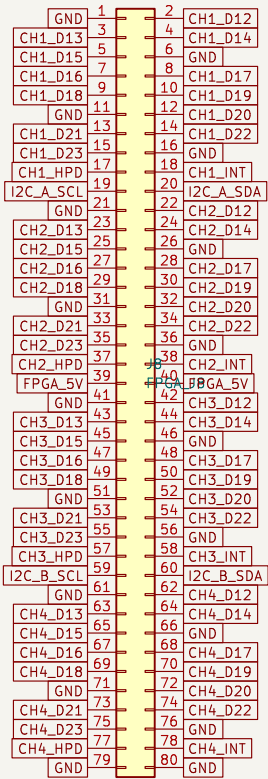
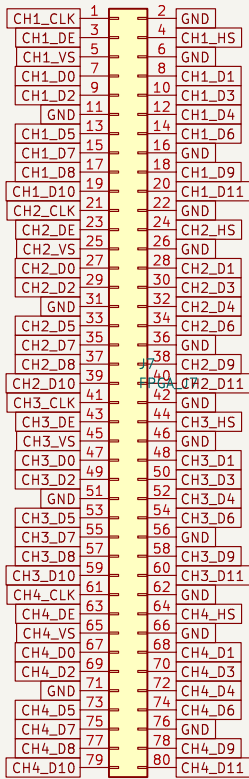


Alternative 5 V input: J6 (2.54 mm header).



Id: 2/

FPGA interface: 2x 80-pin 2.54 mm SMD headers on the bottom side, parallel to the long edge. Header column k (pins 2k-1/2k) sits under the ADV7513 of OUT ceil(k/10): J7 = CLK/sync/D0-D11, J8 = D12-D23/HPD/INT (+I2C, 5V). 3.3 V LVCMOS, 148.



JP1 closed: this board powers the FPGA board through J8 pins 39-40 (FPGA_5V). Leave open if the FPGA board has its own supply.

Quad HDMI test pattern generator - ADV7513 x4 carrier

gyuhooo

Sheet: /FPGA interface headers/

File: fpga_conn.kicad_sch

Title: **FPGA interface headers**

Size: A3

Date: 2026-09-14

Rev: **A**

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Id: 3/7

HDMI OUT1 I2C bus A, ADV7513 address 0x72 (PD/AD=L)

Unused audio inputs (SPDIF, MCLK, I2S0-3, SCLK, LRCLK) tied to GND. CEC_CLK: OR to GND (lift to add a 3-100 MHz clock).

Decoupling: place 0.1uF at each supply pin. C_5V_OUT near TPD12S016 pin 13. AVDD/PVDD/BGVDG feed through ferrite beads.

Quad HDMI test pattern generator - ADV7513 x4 carrier
 gyuhooo
 Sheet: /HDMI OUT1: ADV7513 + TPD12S016/
 File: hdmi_ch1.kicad_sch
Title: HDMI OUT1: ADV7513 + TPD12S016
 Size: A3 Date: 2026-09-14 Rev: A
 KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 Id: 4/7

HDMI OUT1 I2C bus A, ADV7513 address 0x72 (PD/AD=L)

Unused audio inputs (SPDIF, MCLK, I2S0-3, SCLK, LRCLK) tied to GND. CEC_CLK: OR to GND (lift to add a 3-100 MHz clock).

Decoupling: place 0.1uF at each supply pin. C_5V_OUT near TPD12S016 pin 13. AVDD/PVDD/BGVDG feed through ferrite beads.

Quad HDMI test pattern generator - ADV7513 x4 carrier
 gyuhooo
 Sheet: /HDMI OUT1: ADV7513 + TPD12S016/
 File: hdmi_ch1.kicad_sch
Title: HDMI OUT1: ADV7513 + TPD12S016
 Size: A3 Date: 2026-09-14 Rev: A
 KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 Id: 4/7

HDMI OUT1 I2C bus A, ADV7513 address 0x72 (PD/AD=L)

Unused audio inputs (SPDIF, MCLK, I2S0-3, SCLK, LRCLK) tied to GND. CEC_CLK: OR to GND (lift to add a 3-100 MHz clock).

TMDS pairs: 100 ohm differential, length-matched within 0.1 mm per pair, TPD12S016 within 10 mm of the connector.

Decoupling: place 0.1uF at each supply pin. C_5V_OUT near TPD12S016 pin 13. AVDD/PVDD/BGVDG feed through ferrite beads.

Quad HDMI test pattern generator - ADV7513 x4 carrier
gyuhooo
Sheet: /HDMI OUT1: ADV7513 + TPD12S016/
File: hdmi_ch1.kicad_sch
Title: HDMI OUT1: ADV7513 + TPD12S016
Size: A3 Date: 2026-09-14 Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 Id: 4/7

Quad HDMI test pattern generator – ADV7513 x4 carrier
gyuhooo
 Sheet: /HDMI OUT1: ADV7513 + TPD12S016/
 File: hdmi_ch1.kicad_sch

Title: HDMI OUT1: ADV7513 + TPD12S016		
Size: A3	Date: 2026-09-14	Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/7

Quad HDMI test pattern generator – ADV7513 x4 carrier
gyuhooo
 Sheet: /HDMI OUT1: ADV7513 + TPD12S016/
 File: hdmi_ch1.kicad_sch

Title: HDMI OUT1: ADV7513 + TPD12S016		
Size: A3	Date: 2026-09-14	Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/7

Quad HDMI test pattern generator – ADV7513 x4 carrier
gyuhooo
 Sheet: /HDMI OUT1: ADV7513 + TPD12S016/
 File: hdmi_ch1.kicad_sch

Title: HDMI OUT1: ADV7513 + TPD12S016		
Size: A3	Date: 2026-09-14	Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/7

Quad HDMI test pattern generator – ADV7513 x4 carrier
gyuhooo
 Sheet: /HDMI OUT1: ADV7513 + TPD12S016/
 File: hdmi_ch1.kicad_sch

Title: HDMI OUT1: ADV7513 + TPD12S016		
Size: A3	Date: 2026-09-14	Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/7

HDMI OUT2 I2C bus A, ADV7513 address 0x7A (PD/AD=H)

Unused audio inputs (SPDIF, MCLK, I2S0-3, SCLK, LRCLK) tied to GND. CEC_CLK: OR to GND (lift to add a 3-100 MHz clock).

TMDs pairs: 100 ohm differential, length-matched within 0.1 mm per pair, TPD12S016 within 10 mm of the connector.

Decoupling: place 0.1uF at each supply pin. C_5V_OUT near TPD12S016 pin 13. AVDD/PVDD/BGVDG feed through ferrite beads.

Quad HDMI test pattern generator - ADV7513 x4 carrier
 gyuhooo
 Sheet: /HDMI OUT2: ADV7513 + TPD12S016/
 File: hdmi_ch2.kicad_sch
Title: HDMI OUT2: ADV7513 + TPD12S016
 Size: A3 Date: 2026-09-14 Rev: A
 KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 Id: 5/7

Decoupling: place 0.1uF at each supply pin. C_5V_OUT near TPD12S016 pin 13. AVDD/PVDD/BGVDG feed through ferrite beads.

Unused audio inputs (SPDIF, MCLK, I2S0-3, SCLK, LRCLK) tied to GND. CEC_CLK: OR to GND (lift to add a 3-100 MHz clock).

TMDS pairs: 100 ohm differential, length-matched within 0.1 mm per pair, TPD12S016 within 10 mm of the connector.

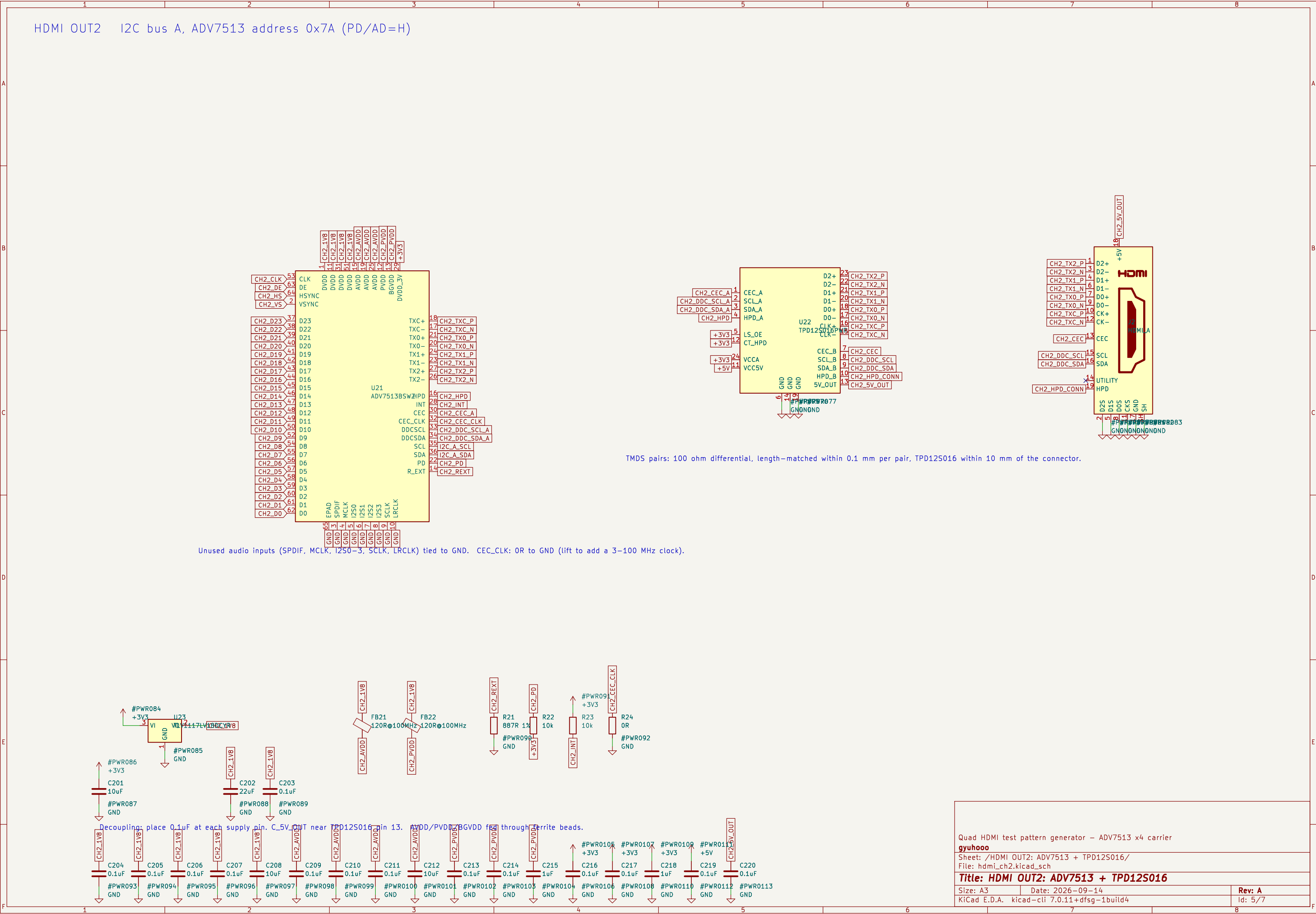
Quad HDMI test pattern generator - ADV7513 x4 carrier
 gyuhooo
 Sheet: /HDMI OUT2: ADV7513 + TPD12S016/
 File: hdmi_ch2.kicad_sch
Title: HDMI OUT2: ADV7513 + TPD12S016
 Size: A3 Date: 2026-09-14 Rev: A
 KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 Id: 5/7

Decoupling: place 0.1uF at each supply pin. C_5V_OUT near TPD12S016 pin 13. AVDD/PVDD/BGVDG feed through ferrite beads.

Unused audio inputs (SPDIF, MCLK, I2S0-3, SCLK, LRCLK) tied to GND. CEC_CLK: OR to GND (lift to add a 3-100 MHz clock).

TMDS pairs: 100 ohm differential, length-matched within 0.1 mm per pair, TPD12S016 within 10 mm of the connector.

Quad HDMI test pattern generator - ADV7513 x4 carrier
 gyuhooo
 Sheet: /HDMI OUT2: ADV7513 + TPD12S016/
 File: hdmi_ch2.kicad_sch
Title: HDMI OUT2: ADV7513 + TPD12S016
 Size: A3 Date: 2026-09-14 Rev: A
 KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 Id: 5/7



HDMI OUT2 I2C bus A, ADV7513 address 0x7A (PD/AD=H)

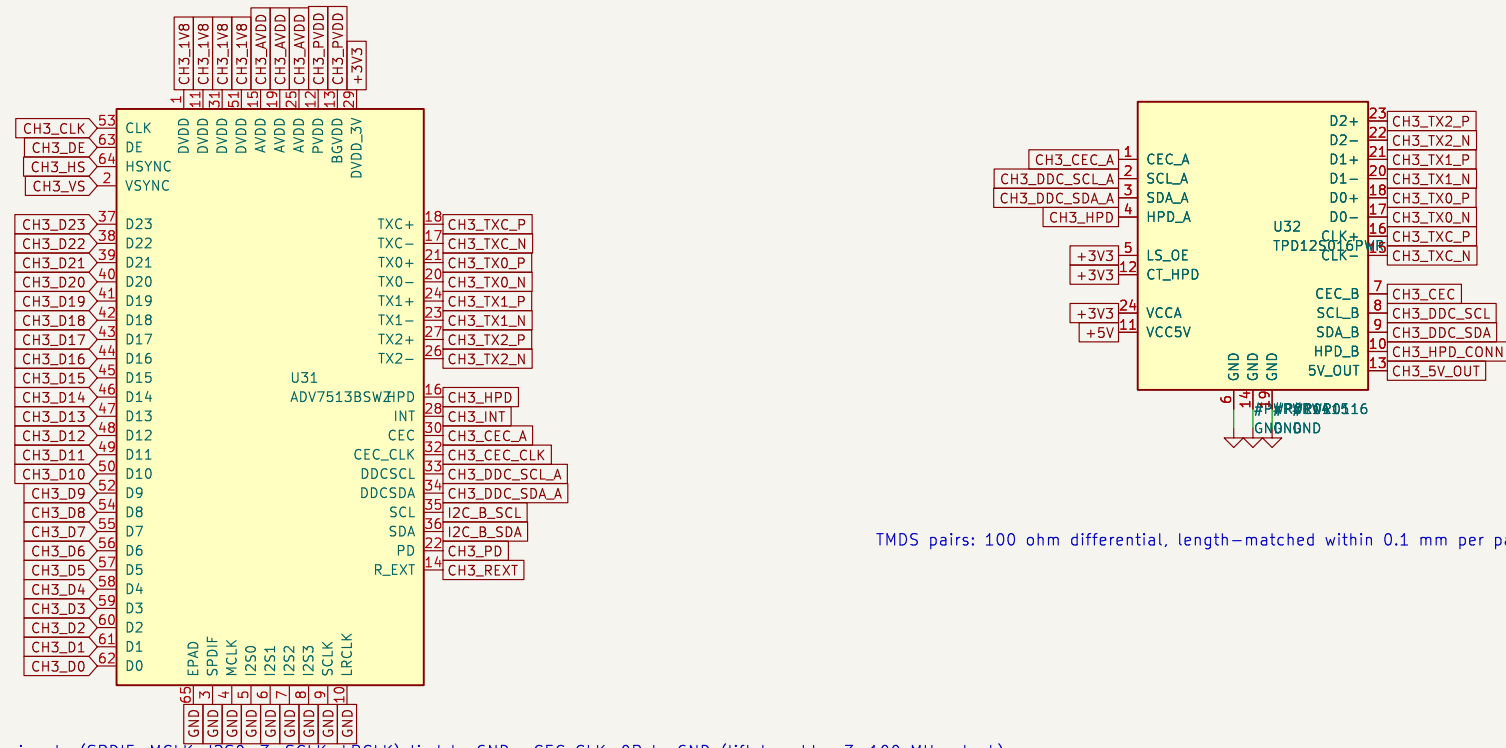
Unused audio inputs (SPDIF, MCLK, I2S0-3, SCLK, LRCLK) tied to GND. CEC_CLK: OR to GND (lift to add a 3-100 MHz clock).

Decoupling: place 0.1uF at each supply pin. C_5V_OUT near TPD12S016 pin 13. AVDD/PVDD/BGVDG feed through ferrite beads.

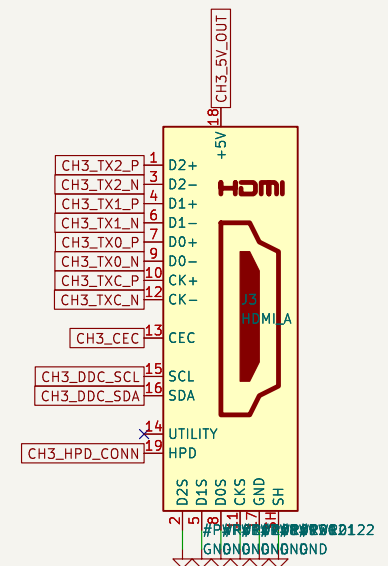
TMDS pairs: 100 ohm differential, length-matched within 0.1 mm per pair, TPD12S016 within 10 mm of the connector.

Quad HDMI test pattern generator - ADV7513 x4 carrier
 gyuhooo
 Sheet: /HDMI OUT2: ADV7513 + TPD12S016/
 File: hdmi_ch2.kicad_sch
Title: HDMI OUT2: ADV7513 + TPD12S016
 Size: A3 Date: 2026-09-14 Rev: A
 KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 Id: 5/7

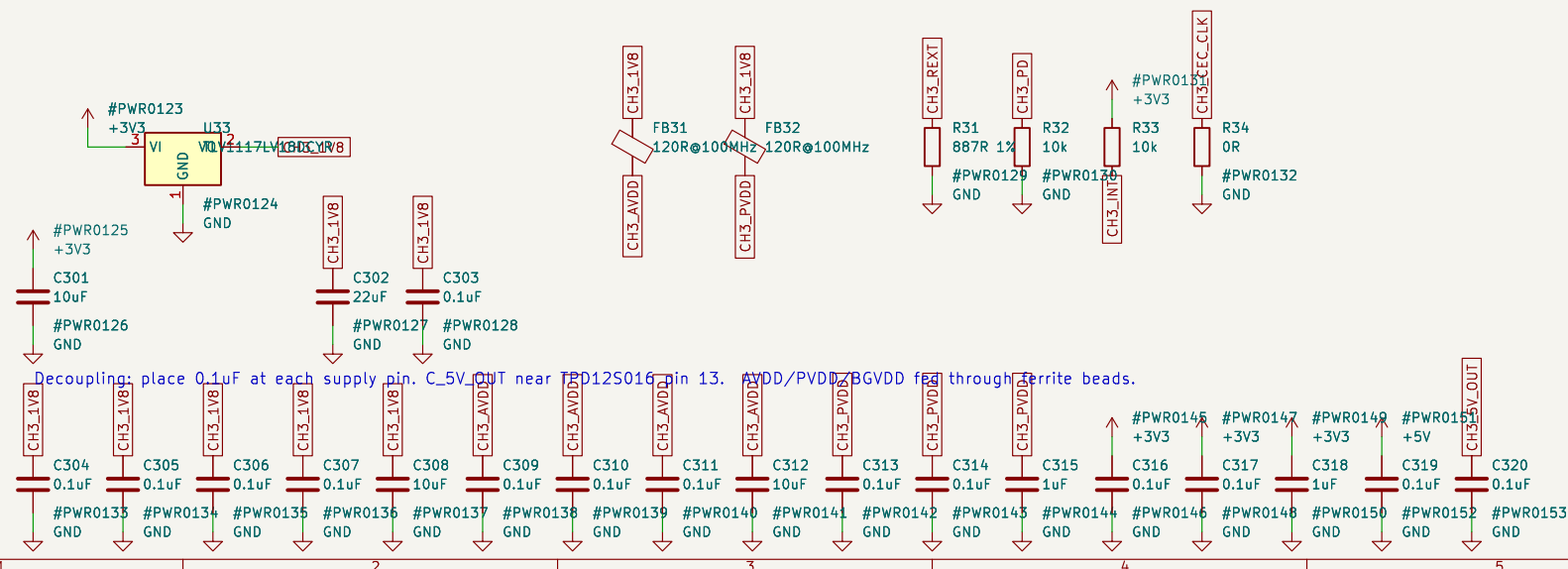
HDMI OUT3 I2C bus B, ADV7513 address 0x72 (PD/AD=L)



TMD5 pairs: 100 ohm differential, length-matched within 0.1 mm per pair, TPD12S016 within 10 mm of the connector.



Unused audio inputs (SPDIF, MCLK, I2S0-3, SCLK, LRCLK) tied to GND. CEC_CLK: OR to GND (lift to add a 3–100 MHz clock).



Quad HDMI test pattern generator – ADV7513 x4 carrier

gyuhooo

Sheet: /HDMI OUT3: ADV7513 + TPD12S016/

File: hdmi_ch3.kicad_sch

Title: HDMI OUT3: ADV7513 + TPD12S016

Size: A3

Date: 2026-09-14

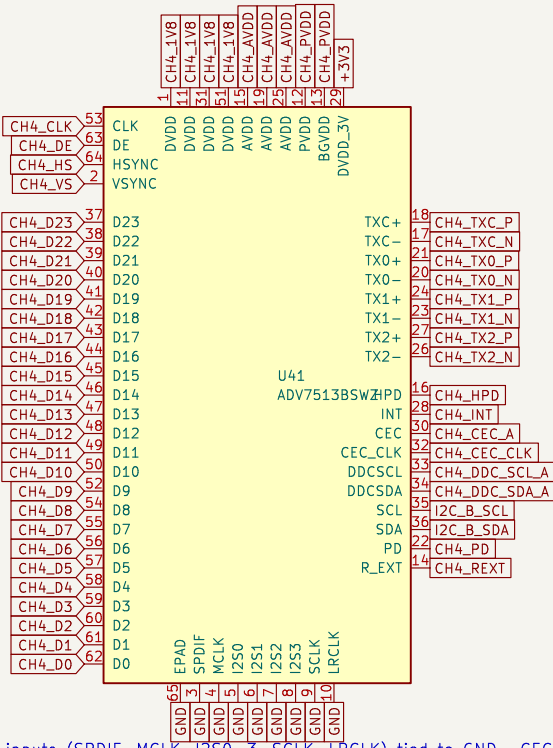
Rev: A

Size: A3	Date: 2020-09-14
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	

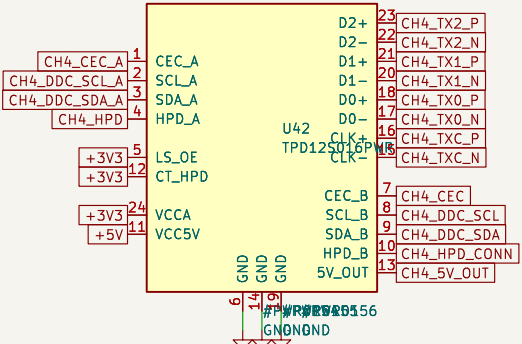
REV: A

Ed: 6/7

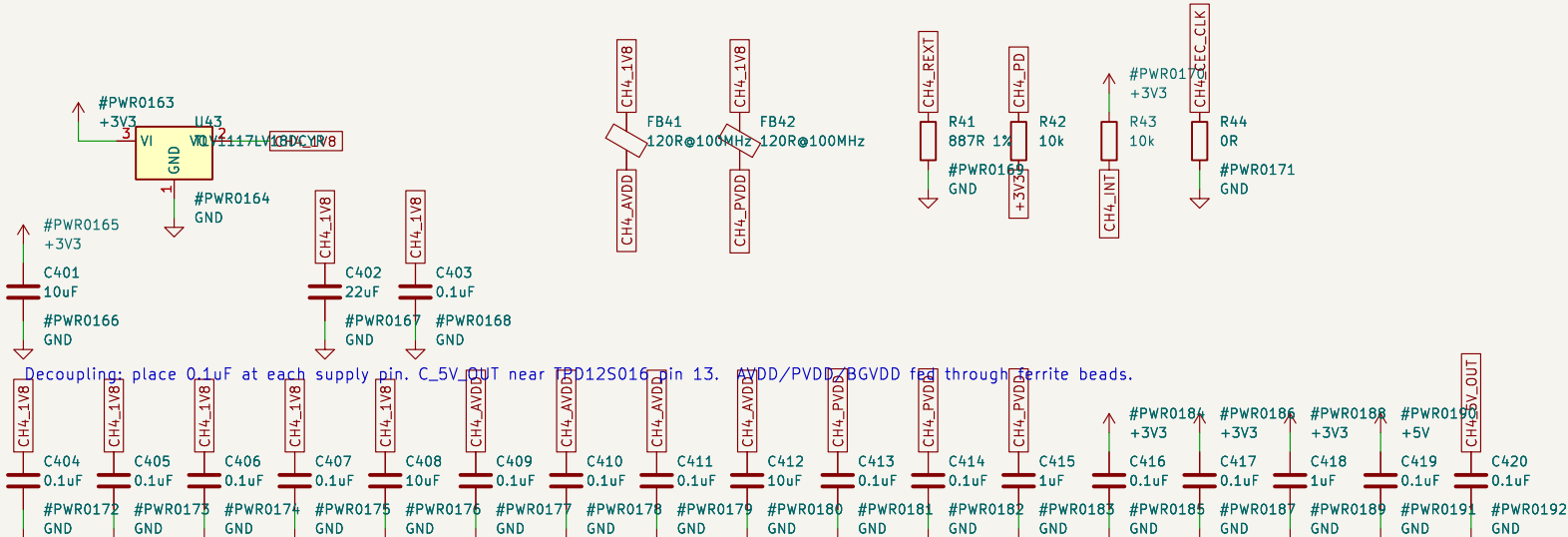
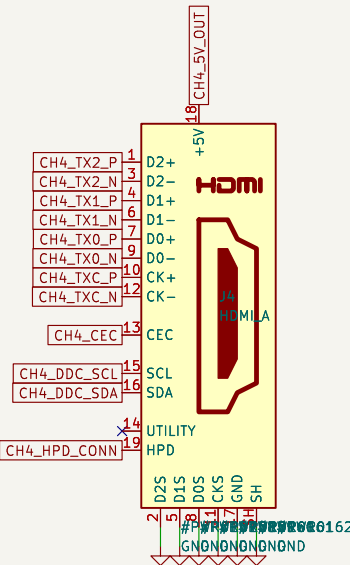
HDMI OUT4 I2C bus B, ADV7513 address 0x7A (PD/AD=H)



Unused audio inputs (SPDIF, MCLK, I2S0-3, SCLK, LRCLK) tied to GND. CEC_CLK: OR to GND (lift to add a 3-100 MHz clock).



TMD5 pairs: 100 ohm differential, length-matched within 0.1 mm per pair, TPD12S016 within 10 mm of the connector.



Quad HDMI test pattern generator – ADV7513 x4 carrier

gyuhooo

Sheet: /HDMI OUT4: ADV7513 + TPD12S016/

File: hdmi_ch4.kicad_sch

Title: HDMI OUT4: ADV7513 + TPD12S016

Size: A3

Date: 2026-09-14

Rev: A

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Id: 7/7